

Observational Data and Nonresponse

STA304 — Summer 2026

Learning Activity 9

Part A: Research Questions and Observational Data

For each research question below, discuss whether it is appropriate to answer using observational data. If it is not fully appropriate, explain what type of conclusion would be more justified.

- a) Does spending more hours studying cause students to earn higher final exam grades?

- b) Are students who live farther from campus less likely to attend lecture regularly?

- c) Does using AI tools improve students' understanding of course material?

- d) Are students who work more hours per week more likely to report feeling overwhelmed?

Part B: Bias from Nonresponse

A survey is conducted to estimate the average number of hours students study per week. The population is divided into two artificial response strata: students who respond to the survey and students who do not respond.

Response Group	Population Size	Mean Study Hours	Response Status
Respondents	600	14	Responds
Nonrespondents	400	8	Does not respond
Total	1000		

- a) Compute the true population mean number of study hours.
- b) Compute the simple respondent average.
- c) Compute the bias of the simple respondent average.
- d) In words, explain why the respondent average is biased in this example.

Part C: Response Propensity

In your own words, define a response propensity.

Part D: Missing Data Mechanisms

For each scenario below, identify whether the missingness is most consistent with MCAR, MAR, or NMAR. Briefly justify your answer.

- a) A random software glitch causes 5% of submitted survey responses to be deleted, regardless of the respondent's answers or characteristics.

- b) In a student wellness survey, younger students are less likely to answer the question about weekly exercise. Age is recorded for all sampled students.

- c) In an income survey, people with very high incomes are less likely to report their income.

- d) In a course feedback survey, students with lower course satisfaction are less likely to complete the survey, and satisfaction is the main outcome of interest.

- e) In a health survey, response rates differ by sex and age group, both of which are available from administrative records for respondents and nonrespondents.